

ZeroHSI: Zero-Shot 4D Human-Scene Interaction by Video Generation

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Abstract

*Human-scene interaction (HSI) generation is crucial for applications in embodied AI, virtual reality, and robotics. Yet, existing methods cannot synthesize interactions in unseen environments such as in-the-wild scenes or reconstructed scenes, as they rely on paired 3D scenes and captured human motion data for training, which are unavailable for unseen environments. We present ZeroHSI, a novel approach that enables **zero-shot** 4D human-scene interaction synthesis, eliminating the need for training on any MoCap data. Our key insight is to distill human-scene interactions from state-of-the-art video generation models, which have been trained on vast amounts of natural human movements and interactions, and use differentiable rendering to reconstruct human-scene interactions. We evaluate ZeroHSI on a curated dataset of different types of various indoor and outdoor scenes with different interaction prompts, demonstrating its ability to generate diverse and contextually appropriate HSIs. Project page: <https://awfuact.github.io/zerohsi>.*

1. Introduction

Generating realistic human motions that interact with 3D scenes is fundamental to computer graphics, VR/AR, embodied AI, and robotics. Despite significant advances in motion synthesis, simulating human-scene interactions that generalize across diverse environments remains a challenge.

Prior work in human-scene interaction synthesis primarily follows two directions: static scene interactions [13, 15, 16] and dynamic object manipulation [21, 22, 26]. Both struggle with generalization and require paired training data. They share a fundamental limitation: *they cannot synthesize interactions in unseen environments* (such as in-the-wild scenes or reconstructed scenes), as they rely on paired 3D scene and motion capture data, which are unavailable for unseen scenarios. This motivates us to address the problem of *zero-shot human-scene interaction*, synthesizing HSIs without training, and eliminating the need for 3D HSI data.

We propose ZeroHSI, which integrates video generation and neural human rendering to synthesize zero-shot 4D human-scene interactions by generating videos and extracting motions via differentiable rendering. ZeroHSI can

synthesize appropriate human motions in both static scenes and environments containing dynamic objects, without requiring any ground truth motion data. Our design leverages pretrained video generation models, maintaining compatibility with various generation approaches and allowing seamless integration of future advancements in video generation. Our primary contributions can be summarized as follows:

- We introduce the novel task of zero-shot HSI motion generation, addressing the fundamental limitation of requiring paired motion-scene training data.
- We propose ZeroHSI, which integrates video generation with differentiable rendering to tackle the challenging task.
- We curate a dataset of various indoor and outdoor scenes with different interaction prompts and plausible initial human poses to evaluate zero-shot HSI. We demonstrate that ZeroHSI can generate diverse and contextually appropriate human-scene interactions across these environments.

2. Related Work

Human-Scene Interaction Synthesis. Research in human-scene interaction has progressed along multiple directions. With paired scene-motion datasets [1, 11, 12, 35, 42] and object-motion data [13, 40], researchers have developed methods [1, 13, 20, 33, 34, 40] for generating scene-aware human motions like sitting and reaching. In the domain of object manipulation, research has evolved from primarily focusing on hand motion synthesis [7, 39, 41] to incorporating full-body motions [4, 9, 23, 29, 36] with the fuel of full-body interaction datasets [8, 28]. Building on paired human-object motion data [3, 21, 32], recent methods focus on predicting interactions from past states [32, 37] or object motion sequences [21]. While these approaches have shown promising results, they typically require paired motion-scene data. In contrast, we focus on zero-shot interaction synthesis, eliminating the need for scene-specific motion capture data.

3. ZeroHSI

Formulation. Our goal is to generate a plausible 3D human-scene interaction motion sequence $\tau = \{(\mathcal{M}_t, \mathbf{P}_t)\}_{t=1}^T$, conditioned on a 3D scene $\mathcal{S}$, an interactable dynamic object $\mathcal{O}$, a text prompt describing the interaction c , and the initial states of human and objects τ_0 , where $\mathcal{M}_t$ represents the human pose in frame t , $\mathbf{P}_t \in \mathbb{R}^6$ represents the object's 6D pose in frame t , and T represents the length of the sequence.

*Equal contribution. Work was done while H. Li was a visiting student at Stanford University. H. Li is now with Peking University.

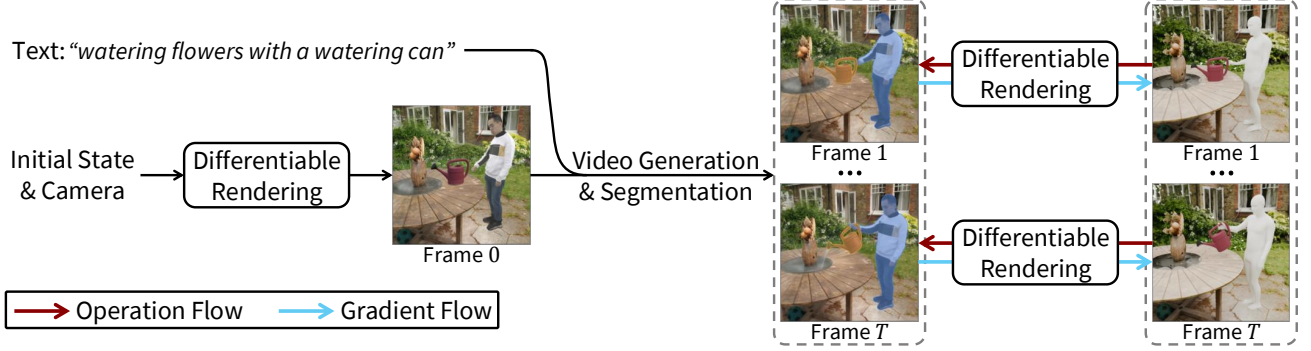


Figure 1. **Overview of ZeroHSI.** Our approach begins with HSI video generation conditioned on the rendered initial state and text prompt. Through differentiable neural rendering, we optimize per-frame camera pose, human pose parameters, and object 6D pose by minimizing the discrepancy between the rendered and generated reference videos.

We use SMPL-X [25], a widely used parameterized human model, to represent the human pose. In each frame, $\mathcal{M}_t$ consists of root translation $\mathbf{r}_t$, global orientation ϕ_t , and body pose Θ_t .

Overview. We present an illustration of our zero-shot human-scene interaction motion generation method in Fig. 1. To address the diversity and generalizability issues encountered by most existing learning-based methods, we first generate HSI video using off-the-shelf video generation models [30] (Sec. 3.2). We then reconstruct the generated 2D HSI video into 4D human-scene-interaction sequence (Secs. 3.3 and 3.4). In addition, we incorporate a refinement process using human pose priors to improve the results (Sec. 3.5).

3.1. Preliminaries

Animatable Gaussians. Animatable Gaussians [24] is a neural human rendering method that maps SMPL-X parameters [25] to an animatable avatar represented by 3DGS [17]. The appearance of the posed avatar can be differentially rendered to image $\mathbf{I}$ through 3DGS rendering $\mathcal{R}$ given camera pose $\mathbf{T} \in \mathbb{R}^{4 \times 4}$, denoted as $\mathbf{I} = \mathcal{R}(\mathcal{G}_H; \mathbf{T})$. Given the driving SMPL-X pose Θ , Animatable Gaussians first deforms the character-specific template using a linear blend skinning (LBS) function. It then predicts Gaussian maps conditioned on the driving pose Θ and the camera view $\mathbf{T}$:

$$\{\Delta\mu_p\}_{p=1}^P \leftarrow \mathcal{F}_{position}(\Theta), \quad (1)$$

$$\{\alpha_p, \mathbf{s}_p, \mathbf{r}_p\}_{p=1}^P \leftarrow \mathcal{F}_{other}(\Theta), \quad (2)$$

$$\{\mathbf{c}_p\}_{p=1}^P \leftarrow \mathcal{F}_{color}(\Theta, \mathcal{V}(\mathbf{T})), \quad (3)$$

where $\Delta\mu_p$ is the offset position relative to the deformed template, and $\mathcal{V}$ is the view direction feature extractor. The root translation $\mathbf{r}$ and the global orientation ϕ are finally applied to $\{\mu_p, \mathbf{s}_p\}_{p=1}^P$ to achieve global transformation. We denote the mapping from human pose and camera view to Gaussian particles proposed by Animatable Gaussians as

$$\mathcal{G}_H \leftarrow \mathcal{A}(\mathbf{r}, \phi, \Theta; \mathbf{T}). \quad (4)$$

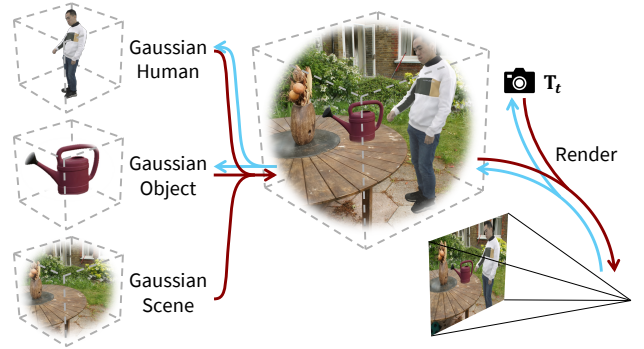


Figure 2. **Illustration of the differentiable rendering process.** The parameterized Gaussian human, transformed Gaussian object, and static Gaussian scene are concatenated and rendered through Gaussian rasterization.

3.2. HSI Video Generation and Processing

Given the initial human pose $\mathcal{M}_0 = (\mathbf{r}_0, \phi_0, \Theta_0)$ and an additional camera pose input $\mathbf{T}_0$, we first map them to the initial human Gaussians $\mathcal{G}_H^0$ through Eq. (4). The initial object 6D pose $\mathbf{P}_0$ is then applied to the Gaussian object $\mathcal{G}_O$, denoted as $\mathcal{G}_O^0 = \mathcal{G}_O(\mathbf{P}_0)$. As depicted in Fig. 2, the resulting Gaussians are concatenated with the Gaussian scene $\mathcal{G}_S$ and rendered into the initial HSI image $\mathbf{I}_0$ under view $\mathbf{T}_0$.

The HSI video $\{\mathbf{I}_t\}_{t=0}^T$ is generated through the KLING image-to-video model [30], conditioned on the initial frame $\mathbf{I}_0$ and text prompt c . We apply Segment Anything Model 2 [27] with automatically selected prompt points to segment the dynamic foreground (human and object) from the static background, obtaining segmentation masks $\{\mathbf{M}_H^t\}_{t=0}^T$ and $\{\mathbf{M}_O^t\}_{t=0}^T$ for human and object, respectively.

3.3. Camera Pose Estimation

We sequentially estimate the relative camera transformation between nearby frames for camera pose estimation. In frame t , we apply a learnable transformation $\mathbf{T} \in \mathbb{R}^{4 \times 4}$ to the Gaussian scene $\mathcal{G}_S$, denoted as $\mathcal{G}_S(\mathbf{T})$. We render image with the estimated camera pose $\mathbf{T}_{t-1}$, and optimize the rel-

ative transformation $\mathbf{T}$ with the photometric loss between frame t , focusing on the static background region:

$$\mathbf{T}_* = \arg \min_{\mathbf{T}} \mathcal{L}_2(\mathcal{R}(\mathcal{G}_S(\mathbf{T}); \mathbf{T}_{t-1}) \odot \mathbf{M}_t, \mathbf{I}_t \odot \mathbf{M}_t), \quad (5)$$

where $\mathbf{M}_t$ represents the mask of static background. The camera pose for frame t is calculated by $\mathbf{T}_t = \mathbf{T}_*^{-1} \mathbf{T}_{t-1}$.

Handling Incorrect Contents. Video generation models produce incorrect contents in regions initially occluded by humans or objects that later become visible (Appendix D.1). Instead of using only the current frame’s static background mask, we address this by aggregating dynamic foreground masks from frames 0 to t to eliminate incorrect contents:

$$\mathbf{M}_t = \mathbf{1} - \bigcup_{i=0}^t (\mathbf{M}_{\mathcal{H}}^i \cup \mathbf{M}_{\mathcal{O}}^i). \quad (6)$$

3.4. HSI Optimization

The optimization of 4D human-scene interactions is carried out sequentially on a frame-by-frame basis. For object 6D pose estimation, we also optimize the relative transformation $\mathbf{P} \in \mathbb{R}^6$ between nearby frames. To handle cumulative errors caused by video generation quality issues such as body part disappearance (Appendix D.2), we directly optimize the human pose parameters $\mathcal{M}_t = (\mathbf{r}_t, \phi_t, \Theta_t)$, where the root translation $\mathbf{r}_t$ and global orientation ϕ_t are initialized using their respective values $\mathbf{r}_{t-1}$ and ϕ_{t-1} from the previous frame. The body poses $\{\Theta_t\}_{t=1}^T$ are initialized using independent frame-wise estimates from SMPLer-X [6].

Similar to Sec. 3.2, we render the Gaussians human, object, and scene at frame t through

$$\begin{aligned} \hat{\mathbf{I}}_t &= \mathcal{R}(\mathcal{G}_{\mathcal{H}}^t, \mathcal{G}_{\mathcal{O}}^t, \mathcal{G}_S; \mathbf{T}_t) \\ &= \mathcal{R}(\mathcal{A}(\mathbf{r}_t, \phi_t, \Theta_t; \mathbf{T}_t), \mathcal{G}_{\mathcal{O}}(\mathbf{P}_t), \mathcal{G}_S; \mathbf{T}_t), \end{aligned} \quad (7)$$

where $\mathbf{P}_t$ denotes the object’s 6D pose in frame t , computed as the composition of the relative transformation $\mathbf{P}$ and the previously optimized object pose $\mathbf{P}_{t-1}$.

Using the generated video as the reference, we optimize the human pose parameters $\mathcal{M}_t$ with the photometric loss in each frame:

$$\mathcal{L}_{\text{rgb}} = (1 - \lambda) \mathcal{L}_1(\hat{\mathbf{I}}_t, \mathbf{I}_t) + \lambda \mathcal{L}_{\text{D-SSIM}}(\hat{\mathbf{I}}_t, \mathbf{I}_t). \quad (8)$$

We introduce two additional loss terms to enhance the accuracy of object 6D pose optimization. For frame t , we compute the center point position $C_{\mathcal{O}}^t \in \mathbb{R}^2$ of the object’s segmentation mask $\mathbf{M}_{\mathcal{O}}^t$ and the center point position of the rendered object region $\hat{C}_{\mathcal{O}}^t$. The object center point position loss is defined as

$$\mathcal{L}_{\text{center}} = \mathcal{L}_2(\hat{C}_{\mathcal{O}}^t, C_{\mathcal{O}}^t). \quad (9)$$

A depth regularization term is incorporated based on the assumption that the object’s depth remains relatively constant throughout a short time window (e.g., 5 seconds). We compute the average object depth in the first frame $D_{\mathcal{O}}^0$ and the average depth of the rendered object region $\hat{D}_{\mathcal{O}}^t$, and define the depth regularization loss as

$$\mathcal{L}_{\text{depth}} = \mathcal{L}_2(\hat{D}_{\mathcal{O}}^t, D_{\mathcal{O}}^0). \quad (10)$$

The object’s relative transformation $\mathbf{P}$ is optimized using the composite loss function:

$$\mathcal{L} = \mathcal{L}_{\text{rgb}} + \lambda_{\text{center}} \mathcal{L}_{\text{center}} + \lambda_{\text{depth}} \mathcal{L}_{\text{depth}}. \quad (11)$$

Handling Appearance Change. Video generation models do not guarantee a consistent human appearance (Appendix D.3), which can potentially compromise human pose optimization. We fine-tune the color net defined in Eq. (3) during optimization using $\mathcal{L}_{\text{rgb}}$.

3.5. Refinement

Limited supervision from single-view video poses challenges for natural human motion reconstruction. We refine our results in the latent space of VPoser [25]. For each frame, the reference joint positions $\hat{J}_t$ are computed by passing the raw human pose parameters obtained through Sec. 3.4 to the SMPL-X layer [25]. We then optimize the root translation $\mathbf{r}_t$, global orientation ϕ_t , and the body pose latent $\mathbf{z}_t \in \mathbb{R}^{32}$ with the fitting loss:

$$\mathcal{L}_{\text{fit}} = \frac{1}{T} \sum_{t=0}^T \mathcal{L}_2(\hat{J}_t, J_t(\mathbf{r}_t, \phi_t, \mathcal{D}(\mathbf{z}_t))), \quad (12)$$

where $\mathcal{D}$ denotes the VPoser decoder. Following prior works [14, 37, 38], we incorporate physics loss $\lambda_{\text{physics}} \mathcal{L}_{\text{physics}}$ to enhance physical plausibility, detailed in Appendix C.3.

4. Experiments

Evaluation Dataset. We introduce AnyInteraction, an evaluation benchmark comprising various 3D environments sourced from existing datasets [16], public asset libraries, and real-world reconstructed scenes from Mip-NeRF 360 [2] and Tanks and Temples [19]. The dataset encompasses 12 distinct 3D environments (7 indoor and 5 outdoor), detailed in Appendix B. We augment these scenes with interactive object models from BlenderKit and 3dsky. Each scene is annotated with 1-3 natural language descriptions of human-scene interactions and corresponding initial positions.

Baselines. We compare our method with LINGO [15] and CHOIS [22]. LINGO [15] generates grasp/put-down actions by designating goal locations for hand-object attachment and release. CHOIS [22] synthesizes human-object interaction guided by language and sparse object waypoints.

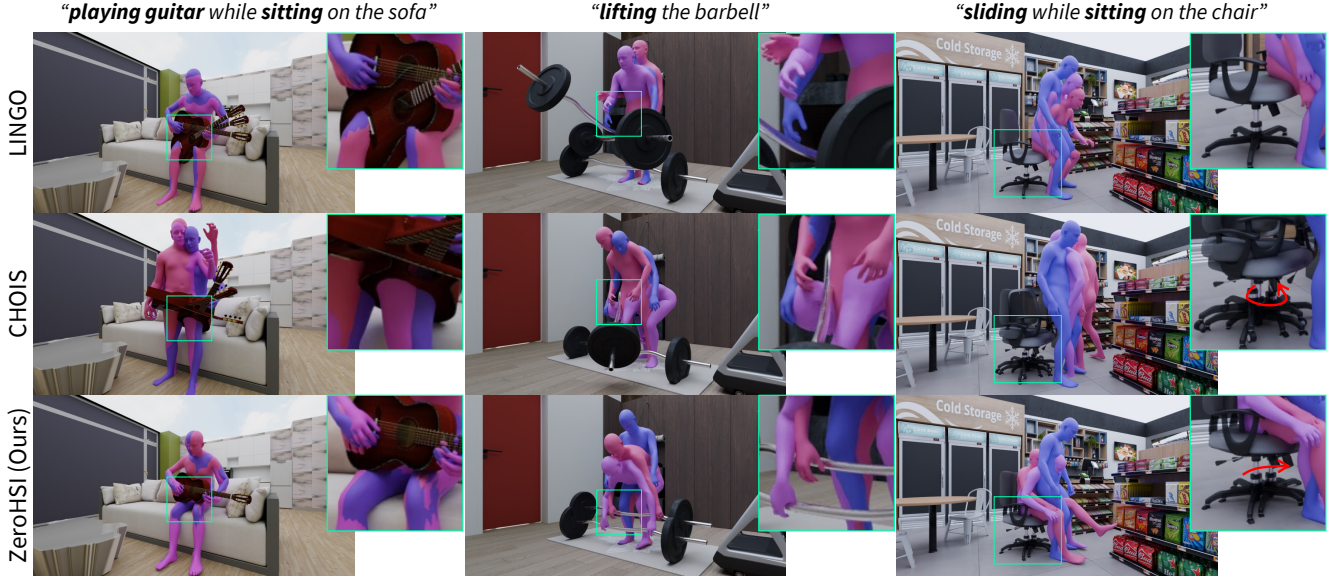


Figure 3. **Qualitative comparison of interactions with dynamic objects in scenes on AnyInteraction.** Our method maintains proper object contact while minimizing penetration, successfully handling challenging interactions like sliding while seated on an office chair.

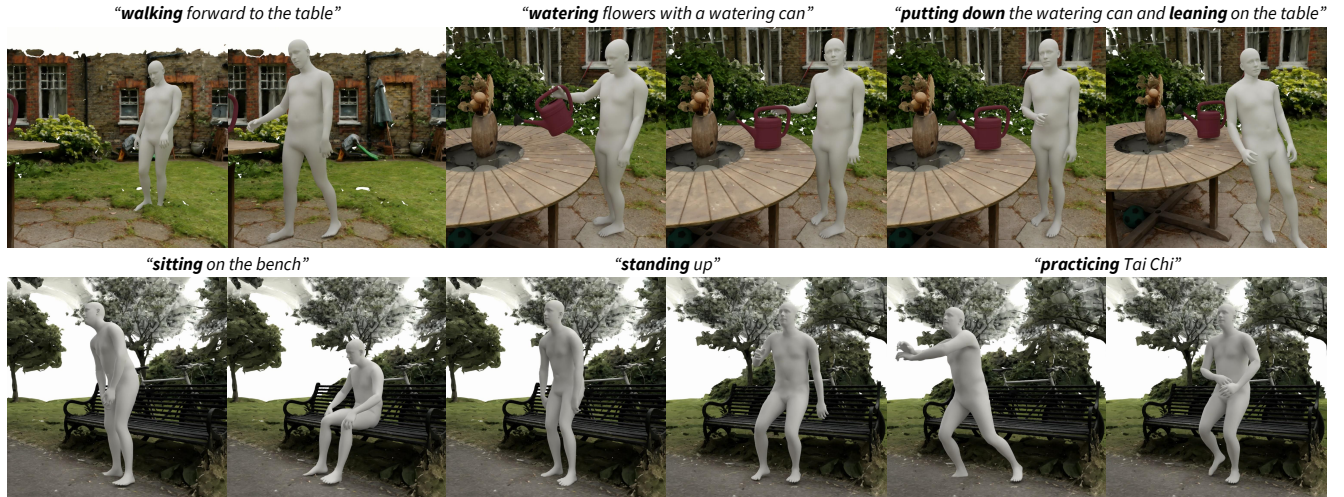


Figure 4. **Qualitative results of long-term interactions with reconstructed real scenes on AnyInteraction.** ZeroHSI generates long-term interaction sequences with multiple text prompts in reconstructed in-the-wild scenes from Mip-NeRF 360 (Garden and Bicycle) dataset [2].

4.1. Comparisons

Our method excels in interactions involving dynamic objects. As illustrated in Fig. 3, while LINGO [15] and CHOIS [22] generate semantically relevant motions, they suffer from object penetration issues (playing guitar) and poor contact quality (lifting barbell). They also struggle with complex interactions requiring multiple body parts and global translation, such as sliding while seated. Our ZeroHSI consistently handles these challenging settings, generating high-quality interactions with dynamic objects.

4.2. Long-Term Interaction Synthesis

We show examples of long-term HSIs in reconstructed real scene generated by ZeroHSI in Fig. 4. The Garden example shows a character walks forward, waters plants, puts down

the watering can, and leans on the table. In the Bicycle example, the character sits on a bench, stands up, and practices Tai Chi. These examples showcase ZeroHSI’s capability to generate realistic interactions in real-world environments and its application for producing sequences of varying lengths. More details and visualizations are in Appendix C.5.

5. Conclusion

We present ZeroHSI, a zero-shot approach to 4D human-scene interaction generation that addresses the limitation of requiring paired motion-scene training data. Our method successfully distills HSIs from video generation models through neural human rendering to synthesize contextually appropriate interactions across diverse environments.

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